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2027 CYCLE

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CUET UG

Economics

The Theory of the firms under perfect competition

Official Syllabus · Course I: Introductory Microeconomics · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Economics syllabus 2026, the current 2027 baseline, Course I)	Skills the questions test
Features and profit maximisation	Features of perfect competition; profit maximization	Recalling the price-taker setup and applying the three profit-maximising conditions (MC = MR, rising MC, price ≥ AVC).
Supply	Supply: market supply, determinants of supply, supply schedule, supply curve, movement along and shifts in supply curve, price elasticity of supply, measurement of price elasticity of supply	Building market supply from firms, sorting movements from shifts, and computing elasticity by the percentage method.
Price determination	Price determination under perfect competition — equilibrium price, effects of shifts in demand and supply	Explaining how equilibrium price emerges from D = S and predicting price-quantity outcomes after demand or supply shifts.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Economics paper follows NCERT Class 12 Economics, and this unit is the supply twin of the demand unit you have already finished — every rule you learned for demand (movement vs shift, elasticity, horizontal summation) reappears here with the signs reversed.

Perfect Competition: The Model Market

Picture the vegetable mandi at 6 a.m. — hundreds of sellers with identical tomatoes, hundreds of buyers who know every quote. No single seller can charge a paisa more, because any buyer can walk two stalls down. That is the **structure** called perfect competition: not aggression, but complete powerlessness of each trader to influence price.

Feature	What it means for the firm
Very many buyers and sellers	Each one's share of the market is negligible — one farmer's entire crop cannot dent the market price
Homogeneous product	Identical units (grade-A wheat is grade-A wheat) — no brand, no premium, no advertising
Free entry and exit	Positive profits pull new firms in; persistent losses push firms out — this disciplines long-run outcomes
Perfect knowledge and factor mobility	Everyone knows all prices; workers and machines move freely to their best use
Firm = price taker	The INDUSTRY sets the price through demand and supply; the firm simply accepts it and chooses only its QUANTITY

- **The price-taker consequence:** the firm's demand curve is a HORIZONTAL line at the market price. Sell 10 units or 10,000 — the price is the same P.
- **Revenue collapse:** because every extra unit sells at exactly P: AR = MR = P — one horizontal line wearing three names.
- **Contrast that examiners love:** a monopolist IS the industry — downward-sloping demand, MR below AR. Perfect competition's flat line is the exact mirror image.

Profit Maximisation: The Three Conditions

The firm does not chase maximum OUTPUT; it chases maximum PROFIT. Marginal thinking finds it: make every unit whose extra cost (MC) is below the extra revenue (MR), and stop the moment they equalise — on the way UP.

THE THREE CONDITIONS OF FIRM EQUILIBRIUM

1. MC = MR — the last unit adds exactly as much cost as revenue.

2. MC must be RISING at that point — this kills the false crossing on MC's falling arm.

3. $P \geq$ AVC in the short run — if price cannot cover average variable cost, produce NOTHING and lose only the fixed cost.

PROFIT TABLE — PRICE ₹12

The firm takes the market price ₹12. Watch profit climb, peak and fall:

Q	TR (₹)	TC (₹)	Profit (₹)	MC (₹)	Decision
0	0	10	−10	—	Fixed loss if idle

Q	TR (₹)	TC (₹)	Profit (₹)	MC (₹)	Decision
1	12	20	−8	10	Produce — narrows the loss
2	24	26	−2	6	Produce — almost even
3	36	30	6	4	Produce — profit begins
4	48	42	10	12	STOP — MC = MR = 12, MC rising
5	60	58	2	16	No — costs overtake
6	72	78	−6	20	No — deep loss

- **Profit peaks at Q4 (₹10).** Exactly there MC = MR (12 = 12) and MC is rising (4 → 12 → 16) — both conditions met.
- **Why not Q5?** The 5th unit costs ₹16 but earns ₹12 — it shaves ₹4 off profit. Marginal logic, not totals, stops the firm.

Supply: The Firm's Other Face

Flip the profit rule and supply falls out: at any price, the firm produces every unit whose MC is below the price — so the firm's supply curve IS the rising arm of its MC curve (above AVC minimum). Supply has the same machinery you mastered for demand, with one twist: the price-quantity relation is DIRECT.

- **Supply = willingness + ability to OFFER** a quantity at each price, per period of time. Desire alone (demand's trap) has a supply twin: capacity alone is not supply.
- **Law of supply, other things equal:** own price rises → quantity supplied RISES. The supply curve slopes UPWARD — the exact opposite of demand's downward slope.
- **Supply schedule and curve:** a price-quantity table; plotted with price on the vertical axis, it climbs left to right.
- **Market supply = horizontal summation:** at each price, ADD every firm's quantity. At ₹20, if firm A offers 40 kg and firm B offers 60 kg, the market offers 100 kg.

MARKET SUPPLY — SUM OF THREE FARMS

Wheat offered at three prices by three farms:

Price (₹/qtl)	Farm A (qtl)	Farm B (qtl)	Farm C (qtl)	Market supply (qtl)
2,200	30	20	10	60
2,400	40	30	15	85
2,600	50	40	20	110

- At every row: market = A + B + C (e.g. 2,400: 40 + 30 + 15 = 85) — quantities add, prices never do.
- Higher price → every farm extends production → market quantity climbs: the market curve slopes upward for the SAME reason each firm's MC curve does.

- **Determinants of supply (the shift forces):**
  - **Input prices:** costlier labour or fertiliser squeezes production → supply falls.
  - **Technology:** improvement cuts costs per unit → supply rises (the biggest long-run lever).
  - **Taxes and subsidies:** a per-unit tax adds cost (supply falls); a subsidy does the reverse.
  - **Price expectations:** expecting higher prices tomorrow, sellers may HOLD BACK stock today.
  - **Number of firms:** entry adds market supply; exit shrinks it.
  - **Natural factors:** monsoon, pests, strikes — especially for agricultural goods.

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Movement vs Shift: The Supply Version

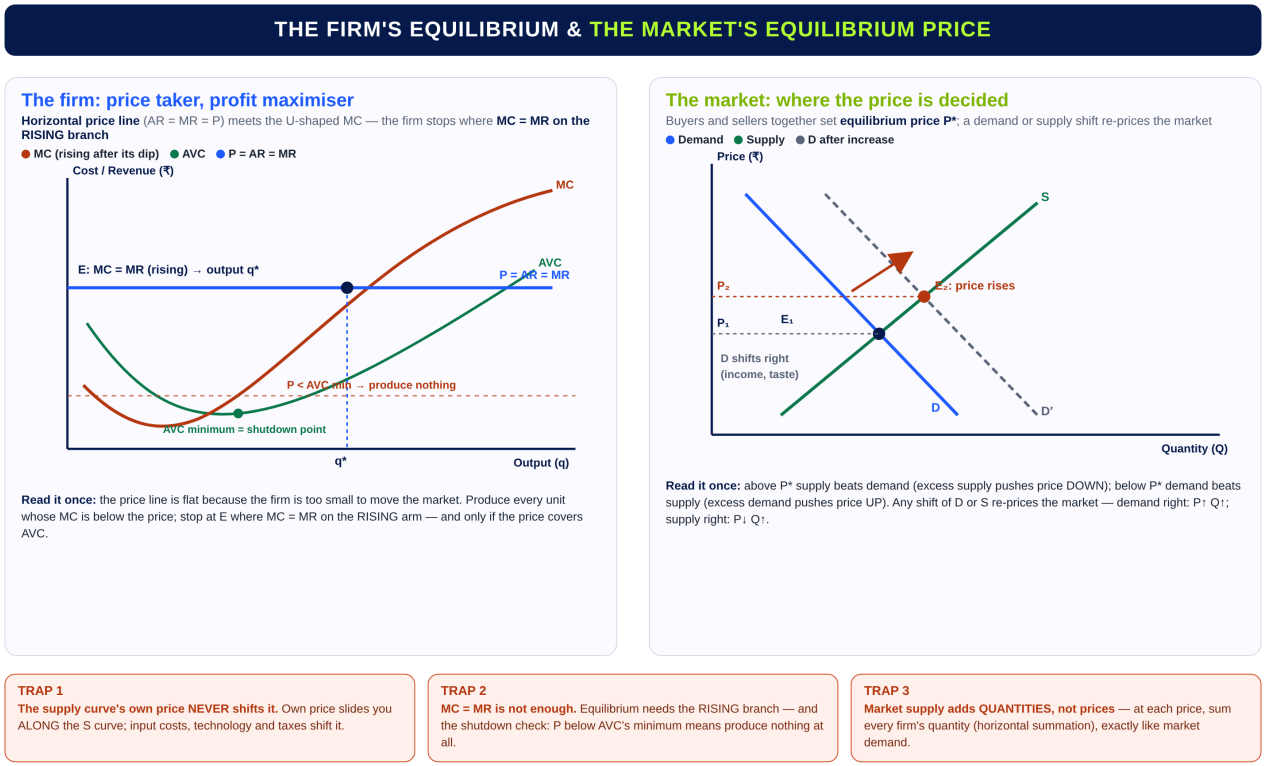


Figure 1: The firm's equilibrium (left): the horizontal price line meets rising MC at E, with the AVC shutdown guard below. The market's equilibrium price (right): D = S fixes P*, and any demand or supply shift re-prices the market.

What changes	Movement or shift?	Direction
The good's OWN price	Movement along S	Price ↑: extension (slide up); Price ↓: contraction (slide down)
Input prices rise	Shift	Supply LEFT (costlier to produce)
Technology improves	Shift	Supply RIGHT
Per-unit tax / subsidy	Shift	Tax: left · Subsidy: right
Price expected to rise	Shift	Hold back today — supply LEFT now
Number of firms rises	Shift	Supply RIGHT

- **Vocabulary, mirrored from demand:** "change in QUANTITY supplied" = movement along the curve (own price); "change in SUPPLY" = a shift (non-price factors). The supply trap is the same trap with an upward-sloping curve.

Price Elasticity of Supply

Same question as demand's elasticity, new curve: when price changes, how much does QUANTITY SUPPLIED respond?

- **Definition:** $e_s = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$ — absolute value; here both move together, so the sign is positive anyway.
- **Percentage method is the syllabus tool** — exactly the four steps you practised for demand:

PES — FULL STEPS

Price of jaggery rises from ₹40 to ₹44 per kg; quantity supplied rises from 500 kg to 600 kg.

- Step 1 — % change in quantity supplied: $(600 - 500)/500 \times 100 = 20\%$
- Step 2 — % change in price: $(44 - 40)/40 \times 100 = 10\%$
- Step 3 — Ratio: $e_s = 20\%/10\% = 2$
- Step 4 — Verdict: $e_s > 1$ — **elastic**: quantity supplied doubled the percentage response of price.

Value	Name	What it looks like
$e_s = 0$	Perfectly inelastic	Fixed quantity however high the price — urban land, a dead artist's paintings
$e_s < 1$	Inelastic	Quantity responds by a SMALLER percentage — perishables, ready stock
$e_s = 1$	Unitary	Both change by the same percentage
$e_s > 1$	Elastic	Quantity responds MORE than proportionally — manufactures with spare capacity
$e_s = \infty$	Perfectly elastic	Unlimited quantity at one price — a single firm's view of demand in perfect competition

- **What makes supply elastic?**
  - **Time period (the biggest lever):** momentary run — fixed supply, $e_s = 0$; short run — variable factors adjust, still inelastic; long run — everything adjusts, most elastic.
  - **Stocks and storage:** large inventories can be released quickly → elastic; perishables → inelastic.
  - **Spare capacity:** idle machines can ramp up fast → elastic.
  - **Nature of the good:** manufactures elastic; agricultural goods inelastic in the short run (the crop is in the ground).

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Price Determination: Where the Price Actually Comes From

Put the market's demand and supply on one diagram and they negotiate. The **equilibrium price** is the only price where the quantity buyers want EXACTLY equals the quantity sellers offer — D crosses S.

- **Above equilibrium:** quantity supplied beats quantity demanded — EXCESS SUPPLY. Unsold stock piles up; sellers undercut each other; price falls toward equilibrium.
- **Below equilibrium:** quantity demanded beats quantity supplied — EXCESS DEMAND. Queues form; buyers bid up; price rises toward equilibrium.
- **Restoring force:** competition among sellers (excess supply) and among buyers (excess demand) walks the price back to P* every time — the market's self-correcting thermostat.

EQUILIBRIUM PRICE — SCHEDULE PROOF

Market demand and supply of eggs (in thousands):

Price (₹/ dozen)	Demand	Supply	Pressure on price
60	70	10	Excess demand 60 — price pushed UP
72	50	30	Excess demand 20 — still rising
78	40	40	D = S — EQUILIBRIUM, no pressure
84	30	50	Excess supply 20 — pushed DOWN
96	10	70	Excess supply 60 — falling hard

- **Equilibrium price: ₹78** — the only price where neither side is frustrated. At ₹60 buyers scramble (shortage); at ₹96 sellers groan (surplus); both mistakes self-correct toward ₹78.
- **Read any row like an auction:** the pressure column IS the market's invisible hand, working row by row.

- **Effects of shifts (memorise the four arrows):**
  - **Demand increases (D right):** P rises, Q rises — festival demand for sweets.
  - **Demand decreases (D left):** P falls, Q falls — a health scare about a food.
  - **Supply increases (S right):** P falls, Q rises — a record monsoon crop.
  - **Supply decreases (S left):** P rises, Q falls — a strike at the factory.
- **Simultaneous shifts:** both increase → Q definitely rises, price DEPENDS on which shift is bigger; both decrease → Q falls, price again ambiguous. When the two moves push price in opposite directions, say "depends on relative magnitude" — that IS the answer.

- **Applications to name-drop:** price supports and price ceilings (next unit's simple applications) bend this same equilibrium — floors above P* create surpluses; ceilings below P* create shortages.

Key Formulas & Rules: One-Stop Revision

PERFECT COMPETITION AND SUPPLY — EVERY RULE ON ONE PAGE

Rule	Statement	When to use it
Price taker	AR = MR = P — horizontal line	Identify the market structure
Firm equilibrium	MC = MR + MC rising + $P \geq$ AVC	All three, or the answer is bait
Shutdown rule	P below AVC's minimum → produce 0	Short-run produce-or-not MCQs
Firm's supply curve	Rising arm of MC above AVC min	Where supply comes from
Law of supply	Own P ↑ → quantity supplied ↑	Upward-sloping S, direct relation
Market supply	Add quantities at each price	Horizontal summation
Supply shifters	Input prices, technology, tax/subsidy, expectations, firm count	Sort movements from shifts
Elasticity of supply	e_s = % change in Q supplied / % change in P	Percentage method, absolute value
Degrees of e_s	0 → less than 1 → 1 → more than 1 → infinity	Perfectly inelastic → perfectly elastic
Time lever	Momentary 0; short run low; long run highest	Rank elasticities by horizon
Equilibrium	D = S; excess supply → P falls; excess demand → P rises	Explain price adjustment
Shift arrows	D right: P↑Q↑ · D left: P↓Q↓ · S right: P↓Q↑ · S left: P↑Q↓	Predict market outcomes

Common Errors & Special Cases

THE TRAPS THIS UNIT SETS — READ BEFORE THE MOCK

The rule you assume	The twist	The trap in the question
Supply slopes down like demand	Supply has a DIRECT relation — it slopes UP; MCQs flip the sign	Curve-identification questions
"Price rose so supply shifted"	Own price NEVER shifts S — it slides you along it	Movement vs shift vocabulary
MC = MR anywhere is equilibrium	Only the RISING-arm crossing counts; the falling crossing is a false stop	Two-crossing MC diagrams
Equilibrium is enough to produce	P below AVC's minimum → shut down even if MC = MR exists	Shutdown-rule numericals
Technology raises price	It cuts costs — supply shifts RIGHT and price FALLS	S-shift direction questions
Tax raises supply	A per-unit tax adds cost — supply shifts LEFT (subsidy: right)	Fiscal lever sorting
PES can be negative	Price and quantity supplied move TOGETHER — e_s is positive by nature	Sign conventions
Both curves shifting gives one price answer	When D and S move oppositely, price "depends on relative magnitude"	Simultaneous-shift MCQs

The 60-Second Self-Test

Cover the answers, say each one aloud, then check. If any answer takes longer than ten seconds, revisit that section.

- **Price rises 15%, quantity supplied rises 9%.** Elasticity? → $e_s = 9/15 = 0.6$ — inelastic.
- **Wages (input cost) jump 20%.** Supply curve moves which way? → LEFT — costlier production at every price.
- **MC = MR at two outputs, MC falling at the first.** Equilibrium? → The SECOND crossing, where MC is rising.
- **Price ₹30, AVC minimum ₹34.** Produce? → No — price cannot cover variable cost; shut down.
- **Two farms offer 25 and 35 units at ₹50.** Market supply at ₹50? → 60 units — horizontal summation.

- **Good harvests everywhere.** Price and quantity of grain? → S shifts right: P falls, Q rises.

The Whole Chapter in One Picture

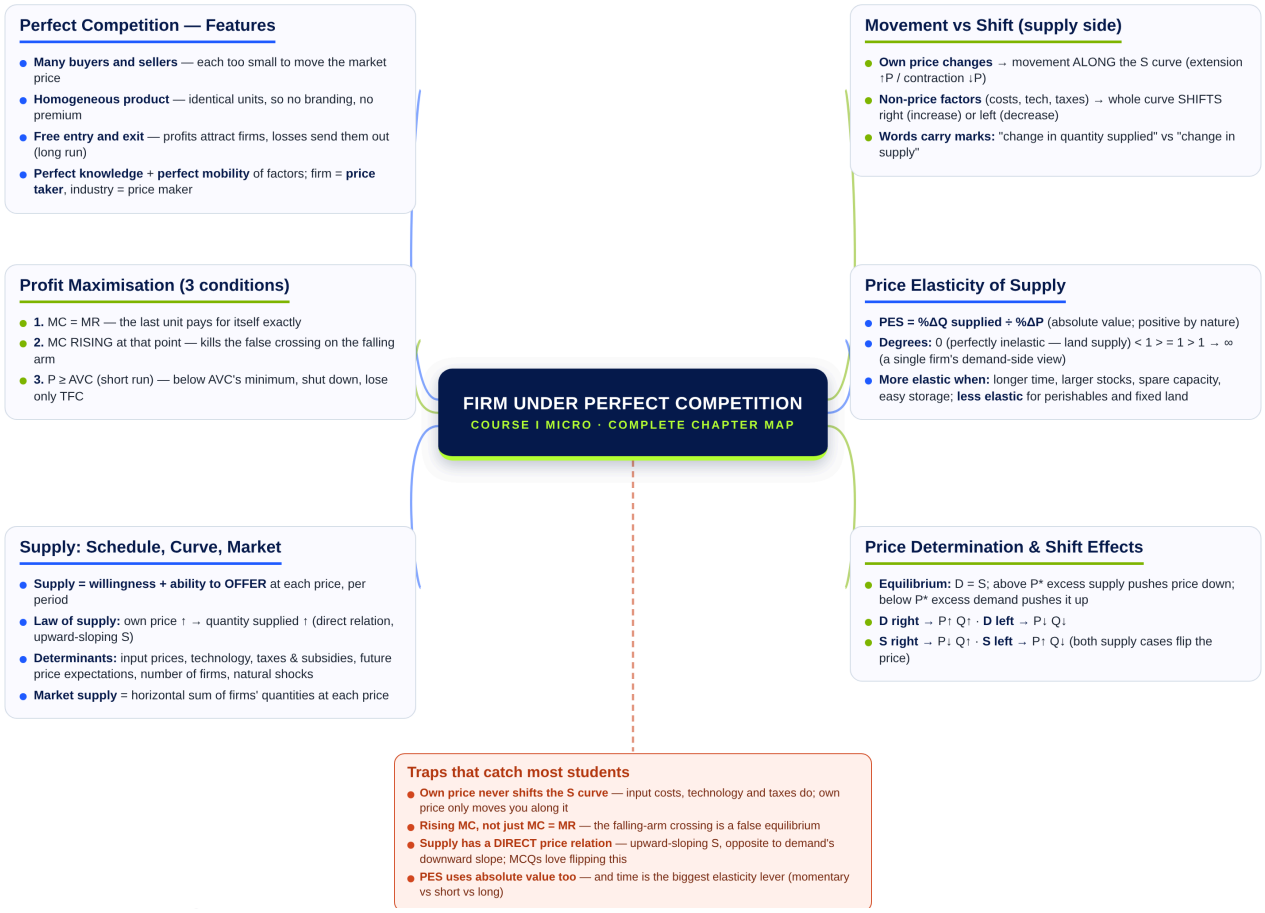


Figure 2: Chapter mindmap: the features of perfect competition, the three profit-maximising conditions, supply's schedule and determinants, movement vs shift, elasticity of supply, and price determination with shift effects — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. Which of the following is NOT a feature of perfect competition?

(A) homogeneous products
(B) free entry and exit of firms
(C) each firm is a price maker
(D) perfect knowledge among buyers

Q2. A firm under perfect competition is a price taker because:

(A) it sells a differentiated product
(B) its share of total output is negligible
(C) the government fixes prices
(D) it faces falling demand

Q3. A firm maximises profit where MC = MR and:

(A) MC is falling
(B) MC is rising
(C) AC is minimum
(D) TR is maximum

Q4. In the short run, a firm should stop production if:

(A) price falls below AVC's minimum
(B) price falls below AC
(C) profit becomes zero
(D) MR starts falling

Q5. Market supply is obtained by:

(A) adding firms' prices at each quantity
(B) adding firms' quantities at each price
(C) averaging the firms' quantities
(D) multiplying price by quantity

Q6. Which of the following would shift the supply curve of steel to the RIGHT?

(A) a rise in iron-ore prices
(B) a new cost-saving furnace technology
(C) a new per-unit tax on steel
(D) expected higher steel prices next month

Q7. A rise in the good's OWN price causes:

(A) a rightward shift of supply
(B) a leftward shift of supply
(C) an upward movement along the same supply curve
(D) a downward movement along the same supply curve

Q8. Price rises by 10% and quantity supplied rises by 25%. The price elasticity of supply is:

(A) 0.4
(B) 2.5
(C) 1
(D) 35

Q9. Supply is MOST elastic for:

(A) urban land
(B) fresh fish caught today
(C) garments with idle machines and big stocks
(D) a monsoon-bound crop already sown

Q10. A bumper monsoon raises the crop supply sharply. In the grain market, equilibrium price will and quantity will :

(A) rise; rise
(B) fall; fall
(C) fall; rise
(D) rise; fall

Detailed Solutions

Each solution shows the working, not just the verdict — cover the answer line, attempt the reasoning, then compare.

Sol. Q1. Perfect competition strips every firm of pricing power: identical products, tiny market shares, free entry, full information. "Each firm is a price maker" describes the opposite structure — a monopoly.

Answer: (C)

Sol. Q2. One farmer out of a lakh cannot move the national wheat price — its output is a drop in the ocean. Negligible market share makes every firm accept the industry price and choose only quantity.

Answer: (B)

Sol. Q3. MC = MR has two crossings on a U-shaped MC. Only the one on the RISING branch stops expansion correctly — before it, the next unit would still add more revenue than cost. AC-minimum and TR-maximum are different events.

Answer: (B)

Sol. Q4. Fixed cost is sunk, but variable cost is avoidable. If price cannot cover even AVC's minimum, every unit produced deepens the loss — shut down and lose only TFC. Falling below AC triggers losses, not shutdown.

Answer: (A)

Sol. Q5. Market supply sums what all firms OFFER at each price — horizontal summation of quantities, exactly parallel to market demand. Prices are never added.

Answer: (B)

Sol. Q6. A cost-saving furnace cuts the cost per tonne, so more steel is profitable at every price — the whole curve shifts RIGHT. Iron-ore prices and taxes shift it left; expected future prices hold back today's stock (left).

Answer: (B)

Sol. Q7. An own-price change never shifts the curve. Price rises → slide UP the same S curve → quantity supplied extends. Shifts belong to costs, technology, taxes and expectations.

Answer: (C)

Sol. Q8. e_s = % change in quantity supplied ÷ % change in price = 25 ÷ 10 = 2.5 — elastic; quantity responded two-and-a-half times as hard as the price.

Answer: (B)

Sol. Q9. Idle machines and large stocks let garment makers ramp up instantly — highly elastic. Land is perfectly inelastic; fresh fish and standing crops cannot wait for a better price.

Answer: (C)

Sol. Q10. A supply increase shifts S right: at the old price there is excess supply, competition among sellers pushes the price DOWN, and the traded quantity RISES. The demand curve never moved.

Answer: (C)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Calling homogeneous products a monopoly trait	Identical products REMOVE brand power — the heart of perfect competition
Q3 option C	Confusing cost events with profit stops	Equilibrium compares MARGINAL cost with MARGINAL revenue — AC's minimum is not the stop rule
Q4 option B	Shutting down on any loss	Produce while P ≥ AVC; losses below AC are survivable short run — only P < AVC forces shutdown
Q7 option A	Shifting on an own-price change	Own price slides you ALONG the S curve; only non-price factors shift it
Q8 option A	Dividing upside down	e_s = %ΔQ ÷ %ΔP = 25 ÷ 10 = 2.5 — quantity first, price second
Q9 option A	Ranking land as elastic	Land is the textbook perfectly inelastic supply — no amount of price adds acres
Q10 option B	Moving both price and quantity down	S right pushes price DOWN but quantity UP — the two always move opposite for supply shifts

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